Stage 1 Hard Drill — With Solutions

Chapter 6.3 — Combinatorial & Two-Player Game Theory

10 new competition-grade questions, full-chapter coverage, full worked solutions. Target: under 90 seconds each once the shortcut is spotted.

Q1 — A Subtraction Game With Period 11

A pile has n stones; a move removes 2, 5 or 6 stones; the player who cannot move loses. Find all P-positions, decide $n = 1000$ and $n = 2025$, find every winning move from 1000, and count the P-positions in $0 \le n \le 100$.

Solution. Build the table: n is P iff all of $n-2, n-5, n-6$ (when ≥ 0) are N.

n	0	1	2	3	4	5	6	7	8	9	10	11
	P	P	N	N	P	N	N	N	P	N	N	P

Here $0, 1$ have no moves; $4 \to 2$ only (N) so P; $8 \to 6, 3, 2$ all N so P; $11 \to 9, 6, 5$ all N so P. Further $12 \to 10, 7, 6$ is P, $13 \to 11$ is N, $14 \to 12$ is N, $15 \to 13, 10, 9$ is P, $16 \to 14, 11$ is N.

Entries $11, \dots, 16$ copy $0, \dots, 5$ (the longest move is 6), so the pattern repeats:

$$P \iff n \equiv 0, 1, 4, 8 \pmod{11}$$

$1000 \equiv 10$: N. Winning moves reach residues $\{0, 1, 4, 8\}$: remove 2 ($\to 998 \equiv 8$) or remove 6 ($\to 994 \equiv 4$); removing 5 gives $995 \equiv 5$ (N).

$2025 \equiv 1$: a P-position, the second player wins.

Counting $0 \le n \le 100$: residue 0: 10 values $(0, \dots, 99)$; residue 1: 10 $(1, \dots, 100)$; residue 4: 9; residue 8: 9:

$$10 + 10 + 9 + 9 = \boxed{38}$$

▶ *Speed trick:* for a subtraction game with largest move k, compute until k consecutive entries repeat an earlier block; that is a proof of periodicity.

Q2 — Bounded Nim: A Twist on XOR

Four heaps; a move removes 1, 2 or 3 stones from *one* heap (never more). Normal play. (a) For $(7, 10, 13, 6)$ decide who wins and list all winning moves. (b) Decide $(5, 9, 12, 4)$ and compare with ordinary Nim.

Solution. Each heap is a subtraction game $\{1, 2, 3\}$ whose Grundy value is $G(n) = \text{mex}\{G(n-1), G(n-2), G(n-3)\} = n \bmod 4$. The sum of games has Grundy value equal to the XOR of components.

(a) $G = (3, 2, 1, 2)$:

$$3 \oplus 2 \oplus 1 \oplus 2 = 2 \neq 0 \;\Rightarrow\; N$$

Target: change one heap's Grundy value g to $g \oplus 2$ by removing 1–3 stones:

$$7\ (g{=}3 \to 1):\ \text{remove } 2 \to 5; \qquad 10\ (2 \to 0) : \to 8$$

$$13\ (1 \to 3) : \to 11; \qquad 6\ (2 \to 0) : \to 4$$

So there are **four** winning moves, each removing 2 stones.

(b) $G = (1, 1, 0, 0)$, XOR $= 0$: a P-position, the *second* player wins. In ordinary Nim $5 \oplus 9 \oplus 12 \oplus 4 = 4 \neq 0$ would say the opposite; the cap on the move size changes the answer.

⋆ *Intuition:* the cap turns a heap of size n into an "effective" Nim heap of size $n \bmod 4$. Only the remainder after groups of four matters.

Q3 — Fibonacci Nim Heaps

Nim with heaps $(13, 21, 34, 55)$. Find the winning move(s). Why does reducing the biggest heap not work?

Solution. In binary:

$$13 = 001101,\ 21 = 010101,\ 34 = 100010,\ 55 = 110111$$

$$001101 \oplus 010101 = 011000$$

$$011000 \oplus 100010 = 111010, \quad 111010 \oplus 110111 = 001101$$

So $s = 001101 = 13$, leading bit 8. Only a heap containing the 8-bit may be reduced:

$$13 = 1101:\ \text{yes}, \qquad 21 = 10101:\ \text{no}$$

$$34 = 100010:\ \text{no}, \qquad 55 = 110111:\ \text{no}$$

The unique winning move is $13 \to 13 \oplus 13 = 0$: **remove the entire heap of** 13, leaving $(21, 34, 55)$ with $21 \oplus 34 = 55$.

For the largest heap, $55 \oplus 13 = 58 > 55$: it would have to *increase*, which is illegal.

▶ *Speed trick:* the winning heap is found by the *leading bit* of s, not by size. Check $a \oplus s < a$ before moving.

Q4 — Misère Nim (Last Stone Loses)

Heaps as given; the player who takes the *last* stone loses. (a) $(1, 3, 5, 7)$. (b) $(1, 1, 1, 4)$. (c) $(1, 1, 1, 1, 5)$. Find the outcome and winning move, and compare each with ordinary Nim.

Solution. Rule: play like normal Nim while at least two heaps would remain larger than 1 after your move. If exactly one heap exceeds 1, reduce it to 0 or 1 so that an **odd** number of 1-heaps remains.

(a) $1 \oplus 3 \oplus 5 \oplus 7 = 0$ with several big heaps: P (the mover loses), as in normal Nim.

(b) One heap exceeds 1 and there are three 1-heaps. Leave an odd number of ones: reduce $4 \to 0$ leaving $(1, 1, 1)$. Opponent takes one, you take one, he takes the last and loses. The *normal* Nim move ($s = 5$, $4 \to 4 \oplus 5 = 1$) leaves $(1, 1, 1, 1)$, an even number of ones: that loses in misère.

(c) Four 1-heaps (even), so reduce $5 \to 1$ leaving five 1-heaps (odd). The normal move $5 \to 0$ leaves four ones, which loses here.

A brute-force search over all moves confirms that the only winning moves are $4 \to 0$ in (b) and $5 \to 1$ in (c).

⋆ *Intuition:* misère Nim is ordinary Nim until the very end, where the endgame of single stones is parity: the last big heap acts as a *tempo switch.*

Q5 — Wythoff's Game: Moves and Counting

Two heaps; a move removes any positive number from one heap, or the same number from both. Normal play. (a) Find all winning moves from $(14, 20)$. (b) How many ordered positions (a, b) with $0 \le a, b \le 30$ are P-positions?

Solution. P-positions are the pairs $(a_k, a_k + k)$ with a_k the smallest unused non-negative integer:

$$(0,0), (1,2), (3,5), (4,7), (6,10), (8,13), (9,15), (11,18),$$

$$(12,20), (14,23), (16,26), (17,28), (19,31), \dots$$

(a) From $(14, 20)$ the position $(14, 20)$ is not a pair. Moves to P-positions:

- remove 2 from the first heap: $(12, 20)$;
- remove 5 from *both*: $(9, 15)$.

Removing from only the second heap would need $(14, b)$ with $b < 20$ to be a pair: none exists (14 pairs with 23).

(b) Pairs with larger entry ≤ 30: $k = 1, \dots, 11$ (up to $(17, 28)$; $(19, 31)$ exceeds 30). Each gives two ordered positions, plus $(0, 0)$:

$$2 \cdot 11 + 1 = \boxed{23}$$

▶ *Speed trick:* memorise the pairs up to about $(17, 28)$ and use $b - a = k$ to check membership instantly: $(9, 15)$ has $b - a = 6$ and $a = 9$ is the sixth "new" number.

Q6 — Grundy's Game (Splitting Heaps)

A move splits one heap into two **unequal** non-empty heaps. Normal play. (a) Tabulate Grundy values for sizes 1–9. (b) For heaps $(7, 9, 5)$ decide the winner and list all winning moves.

Solution. (a) Heaps of size 1, 2 cannot be split, so $G = 0$. Otherwise $G(n) = \text{mex}\{G(a) \oplus G(n-a)\}$ over $a < n - a$:

$$G(3):\ (1,2) \to 0,\ \text{so } 1$$

$$G(4):\ (1,3) \to 1,\ \text{so } 0$$

$$G(5):\ (1,4)\to 0,\ (2,3)\to 1,\ \text{so } 2$$
$$G(6):\ (1,5)\to 2,\ (2,4)\to 0,\ \text{so } 1$$
$$G(7):\ (1,6)\to 1,\ (2,5)\to 2,\ (3,4)\to 1,\ \text{so } 0$$
$$G(8):\ \{0,1,3\}\to 2, \qquad G(9):\ \{2,0,0,2\}\to 1$$

n	1	2	3	4	5	6	7	8	9
G	0	0	1	0	2	1	0	2	1

(*b*) $G(7)\oplus G(9)\oplus G(5)=0\oplus 1\oplus 2=3\neq 0$: first player wins. A split of heap h into (a,b) must satisfy $G(a)\oplus G(b)=G(h)\oplus 3$:

$$9\to(1,8):\ 0\oplus 2=2=1\oplus 3\ \checkmark$$

$$9\to(4,5):\ 0\oplus 2=2\ \checkmark \qquad 5\to(2,3):\ 0\oplus 1=1=2\oplus 3\ \checkmark$$

Heap 7 needs $G(a)\oplus G(b)=3$: its splits give $1,2,1$, none. So the winning moves are exactly $9\to 1{+}8$, $9\to 4{+}5$ and $5\to 2{+}3$ (confirmed by exhaustive search).

⋆ *Intuition:* Sprague–Grundy lets you analyse a splitting game as if it were Nim: each heap hides a Nim-heap of size $G(n)$.

Q7 — A Parametric 2×2 Game

$$A(x)=\begin{pmatrix} x & 2\\ 3 & 1\end{pmatrix}$$

Row maximises. For which x is there a saddle point? Find the value as a function of x and solve the case $x=0$.

Solution. Column 2 has maximum 2 (entries $2,1$). Row 1 has minimum $\min(x,2)$. A saddle at (R_1,C_2) needs $\min(x,2)=2$:

$$x\geq 2:\ \text{saddle at } (R_1,C_2),\ v=2$$

For $x<2$ no saddle. Use $a=x,b=2,c=3,d=1$ with $a+d-b-c=x-4$:

$$v=\frac{ad-bc}{a+d-b-c}=\frac{x-6}{x-4}=\frac{6-x}{4-x}$$

$$p(R_1)=\frac{d-c}{x-4}=\frac{2}{4-x}, \qquad q(C_1)=\frac{d-b}{x-4}=\frac{1}{4-x}$$

Both lie in $(0,1)$ for every $x<2$. At $x=2$: $v=\frac{4}{2}=2$, so the two regimes join continuously.

$x=0$: $p=\frac12$, $q=\frac14$, $v=\frac32$. Check: Row mixing $(\frac12,\frac12)$ gets $\frac32$ against C_1 $(0,3)$ and $\frac32$ against C_2 $(2,1)$. As $x\to-\infty$, $v\to 1$.

▶ *Speed trick:* when a parameter appears, find the saddle threshold first (compare $\underline{v}$ and $\overline{v}$), then apply the 2×2 formula only below it.

Q8 — Hide the Treasure

A hider puts a treasure in one of three boxes worth $1,2,3$ coins. The seeker opens exactly one box; if it is the right one the seeker gets that box's value from the hider, else nothing. Find the value and both optimal strategies. Repeat for values $1,2,4$.

Solution. Payoff to the seeker is the diagonal matrix $\mathrm{diag}(1,2,3)$ (rows = hider's box, columns = seeker's box). If the seeker opens box i with probability y_i, the hider's box i costs v_iy_i. The seeker equalises:

$$v_iy_i=V\ \Rightarrow\ y_i=\frac{V}{v_i}, \qquad \sum y_i=1\ \Rightarrow\ V=\frac{1}{\sum 1/v_i}$$

$$V=\frac{1}{1+\frac12+\frac13}=\boxed{\frac{6}{11}}, \qquad y=\left(\tfrac{6}{11},\tfrac{3}{11},\tfrac{2}{11}\right)$$

The same indifference for the hider gives the same vector $x=y$ and the same value.

Values $1,2,4$: $V=\frac{1}{1+1/2+1/4}=\boxed{\frac47}$, $x=y=\left(\frac47,\frac27,\frac17\right)$.

⋆ *Intuition:* the hider hides in the *cheap* box most often (the valuable box is risky to use), and the seeker mirrors it. Frequencies are inversely proportional to value.

Q9 — The Bluffing Game With a Bet Size b

Each player antes 1. Player 1 is dealt High or Low (probability $\frac12$ each, secret). He may give up (loses 1) or bet b more. Player 2 then folds (Player 1 wins the ante) or calls (Player 1 wins the pot iff High). Strategies: Honest (bet only with High) or Bluff (always bet). Find the value for general b; the case $b=3$; and what happens if Player 2 always calls.

Solution. Payoffs to Player 1 (average over the card):

$$\text{Honest, Call}:\ \tfrac12(1+b)+\tfrac12(-1)=\tfrac{b}{2}$$
$$\text{Honest, Fold}:\ \tfrac12(1)-\tfrac12=0$$
$$\text{Bluff, Call}:\ \tfrac12(1+b)-\tfrac12(1+b)=0$$
$$\text{Bluff, Fold}:\ \tfrac12(1)+\tfrac12(1)=1$$
$$A=\begin{pmatrix} b/2 & 0\\ 0 & 1\end{pmatrix}$$

The denominator of the 2×2 formulas is $\frac{b}{2}+1-0-0=\frac{b+2}{2}$:

$$p(\text{Honest})=q(\text{Call})=\frac{1}{(b+2)/2}=\frac{2}{b+2}$$

$$v=\frac{b/2}{(b+2)/2}=\boxed{\frac{b}{b+2}}$$

$b=3$: $v=\frac35$, Honest with probability $\frac25$ (so a Low card is bet $\frac35$ of the time) and Player 2 calls with probability $\frac25$.

If Player 2 *always calls* (ignoring the equilibrium), Player 1's best reply is Honest, earning $\frac{b}{2}=\frac32$, far more than $\frac35$. If Player 2 always folds, Bluff earns 1.

As $b\to\infty$ the value tends to 1 (big bets make the bluff frequency $\frac{b}{b+2}\to 1$).

▶ *Speed trick:* when the matrix is diagonal, $\begin{pmatrix}\alpha & 0\\ 0 & \beta\end{pmatrix}$ with $\alpha,\beta>0$, the value is $\frac{\alpha\beta}{\alpha+\beta}$ and strategies are $\propto(\beta,\alpha)$.

Q10 — Capstone: A Random Nim Position

Three Nim heaps a,b,c are independent and uniform on $\{1,\dots,7\}$. Find (*a*) the probability the second player wins; (*b*) the expected number of winning moves, over all positions and given that the first player wins; (*c*) the distribution of the number of winning moves; (*d*) the probability that the second player wins with *four* such heaps.

Solution. (*a*) $P\iff c=a\oplus b$ with $a\neq b$ (so $c\geq 1$) and $a\oplus b\leq 7$ automatically:

$$\#P=7\cdot 6=42, \qquad \mathbb{P}=\frac{42}{343}=\frac{6}{49}$$

(*b*) In an N-position, a heap is a winning candidate iff it contains the leading bit of s. The column of that bit has *odd* parity, so an odd number of heaps contain it: 1 or 3 winning moves. Exhaustive count over the 301 N-positions:

189 positions with 1 winning move, 112 positions with 3

$$\text{total}=189+3\cdot 112=525$$
$$\mathbb{E}[\#]=\frac{525}{343}=\frac{75}{49}\approx 1.53, \qquad \mathbb{E}[\#\mid N]=\frac{525}{301}=\frac{75}{43}\approx 1.74$$

(*c*) Exactly 1 with probability $\frac{189}{343}$, exactly 3 with probability $\frac{112}{343}$, and 0 (a P-position) with probability $\frac{42}{343}$; two winning moves never occur.

(*d*) Choose a,b,c freely (343 ways); the fourth heap is forced $d=a\oplus b\oplus c$ and must lie in $\{1,\dots,7\}$, i.e. $a\oplus b\oplus c\neq 0$, which holds in $343-42=301$ cases:

$$\mathbb{P}=\frac{301}{2401}=\frac{43}{343}\approx 0.125$$

slightly above the $\frac{42}{343}$ of three heaps.

⋆ *Full-circle intuition:* XOR parity explains *everything* here: the second player's winning set is a one-in-eight slab, and the leading-bit column's odd parity forces the winning-move count to be 1 or 3.